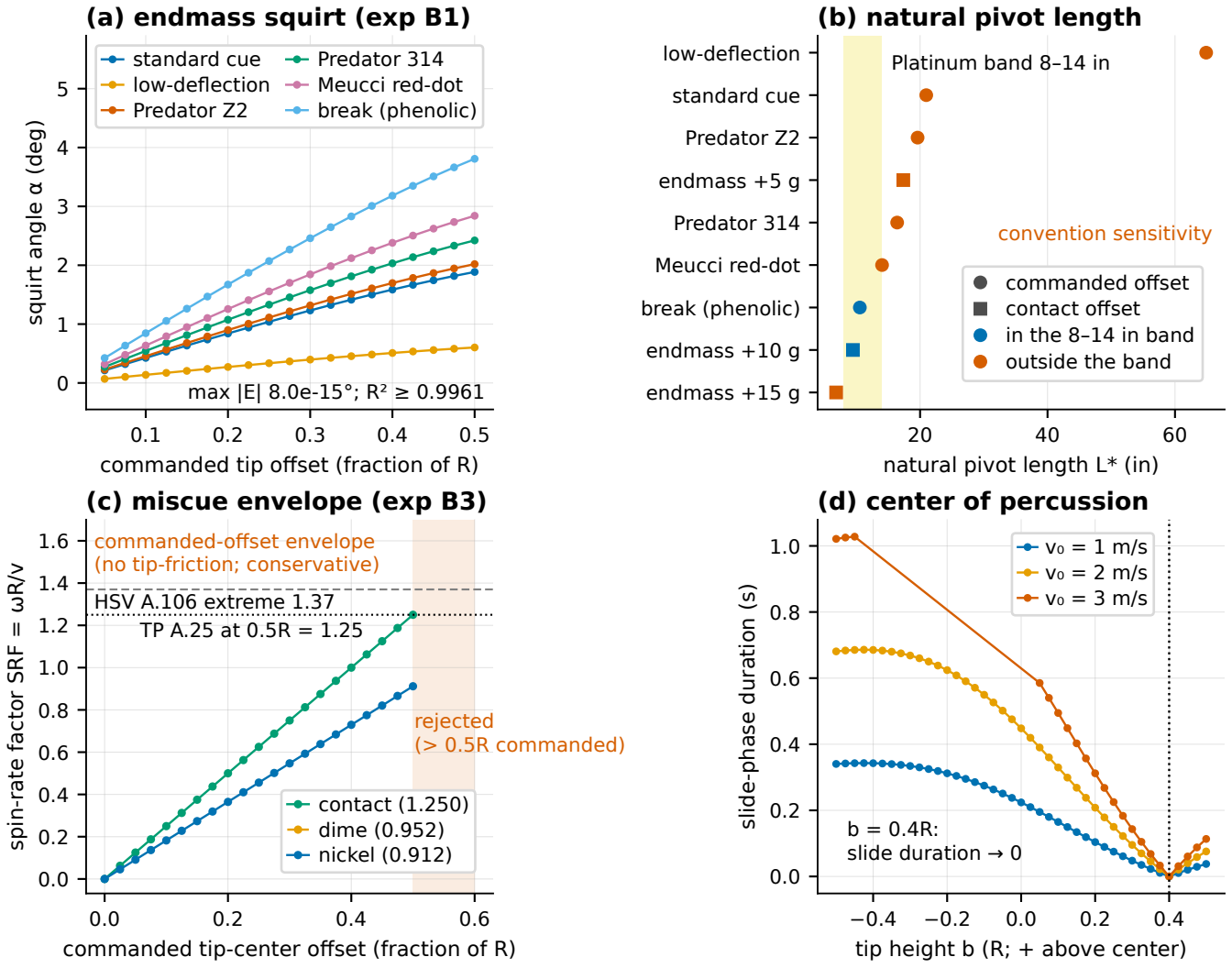
(a) endmass squirt (exp B1)
standard cue
low-deflection
Predator Z2
Predator 314
Meucci red-dot
break (phenolic)
squirt angle α (deg)
max |E| 8.0e-15°; R² ≥ 0.9961
commanded tip offset (fraction of R)
(b) natural pivot length
low-deflection
standard cue
Predator Z2
endmass +5 g
Predator 314
Meucci red-dot
break (phenolic)
endmass +10 g
endmass +15 g
Platinum band 8–14 in
convention sensitivity
commanded offset
contact offset
in the 8–14 in band
outside the band
natural pivot length L* (in)
(c) miscue envelope (exp B3)
commanded-offset envelope
(no tip-friction; conservative)
HSV A.106 extreme 1.37
TP A.25 at 0.5R = 1.25
rejected
(> 0.5R commanded)
contact (1.250)
dime (0.952)
nickel (0.912)
spin-rate factor SRF = ωR/v
commanded tip-center offset (fraction of R)
(d) center of percussion
v₀ = 1 m/s
v₀ = 2 m/s
v₀ = 3 m/s
b = 0.4R:
slide duration → 0
slide-phase duration (s)
tip height b (R; + above center)